Mitigating Overfitting

1) Add a penalty on the coefficients w in the objective function.

This is known Regularization.

$$J(w) = \frac{1}{N}\sum_{i=1}^{N}(t_i - y_i)^2 + \lambda \cdot \begin{cases} \sum_{j=1}^{M} w_j^2 & \text{RIDGE regularizer} \\ \sum_{j=1}^{M} |w_j| & \text{LASSO regularizer} \\ \alpha \cdot \sum_{j=1}^{M} w_j^2 + (1-\alpha) \cdot \sum_{j=1}^{M} |w_j| & \text{ELASTIC NET} \end{cases}$$

hyperparameter

it controls trade-off between minimizing MSE and penalizing coefficients.

$$\lambda \to \infty \therefore \vec{w} = \vec{0}$$

MSE + RIDGE Regularizer

$$\boxed{J(w)} = \frac{1}{N}\sum_{i=1}^{N}(t_i - y_i)^2 + \lambda \cdot \sum_j w_j^2$$

$$= \frac{1}{N}\sum_{i=1}^{N}(t_i - y_i)^2 + \boxed{\frac{\lambda_2}{N}} \sum_j w_j^2$$

λ

$$= \frac{1}{N}\|t - Xw\|^2 + \frac{\lambda_2}{N} \cdot \|w\|^2$$

$$= \frac{1}{N}(t - Xw)^T(t - Xw) + \frac{\lambda_2}{N} w^T w$$

$$\boxed{= \frac{1}{N}\left[t^T t - t^T X w - w^T X^T t + w^T X^T X w\right] + \frac{\lambda_2}{N} w^T w}$$

$$\frac{\partial J}{\partial w} = 0$$

$$\Leftrightarrow -t^T X - (X^T t)^T + (X^T X w)^T + w^T X^T X + 2 \cdot \lambda_2 w^T = 0$$

$$\Leftrightarrow -t^T X + w^T X^T X + \lambda_2 \cdot w^T = 0$$

$$\Leftrightarrow -X^T t + X^T X w + \lambda_2 \cdot w = 0$$

applying transposes on both sides

$$\Leftrightarrow X^T X w + \underbrace{\lambda_2 I w}_{= \lambda_2 \cdot w} = X^T t$$

$$\Leftrightarrow (\underbrace{X^T X}_{(M+1)\times(M+1)} + \lambda_2 \cdot \underbrace{I}_{(M+1)\times(M+1)}) \cdot w = X^T t$$

Diagonally-Loading $X^T X$ matrix.

$$\Leftrightarrow \boxed{w = (X^T X + \lambda_2 \cdot I)^{-1} \cdot X^T t}$$

NORMAL Equations.